# Task 19 — Analytical Data Pipeline Packaging Report

**Author:** Jeevan N | **Track:** Data Analyst | **Environment:** Python 3 / CLI Runner / Logging

## 1. EXECUTIVE SUMMARY & ARCHITECTURE

**Objective:** Refactor ad-hoc exploratory notebooks into a modular, parameterized, and logged ETL data pipeline for repeatable analytics.
**Architecture:** Modularized Python functions chained through a CLI runner (argparse) with active logging at every execution stage.

## 2. PIPELINE STAGE SPECIFICATION & SAFEGUARDS

| STAGE | FUNCTION NAME | INPUT / PARAMETERS | DATA QUALITY / SAFEGUARD | OUTPUT ARTIFACT |
|---|---|---|---|---|
| 1. Extract | extract_data() | --records (N=1000) | Dynamic seed initialization | Raw DataFrame |
| 2. Validate | validate_data() | Raw DataFrame | Assertions (Non-null, >0 amount) | Validated Stream |
| 3. Transform | transform_data() | Validated Stream | Customer spend aggregation | Aggregated Profile |
| 4. Load | load_data() | --output CSV path | Idempotent file overwrite | pipeline_output.csv |

## 3. PITFALL SAFEGUARDS & VERIFICATION

• **Anti-Fragility:** Replaced loose notebook cells with modular, unit-testable functions.
• **Idempotency Enforced:** Output writing verifies and removes pre-existing target files to eliminate duplicate row append bugs.
• **Inter-Stage Validation:** Built-in assertion checks stop execution immediately if nulls or corrupt amounts are detected.

## 4. DELIVERABLES & REPOSITORY VERIFICATION

• **Pipeline Script:** pipeline_runner.py.
• **Output File:** pipeline_output.csv.
• **GitHub Repository:** Live on https://github.com/JeevanN-07/placemux.git.